

Effects of Real-world Backgrounds on User Interface Color Naming and Matching in Automotive AR HUDs

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ABSTRACT

This study investigated which optical see-through AR user interface colors are most robust across different real-world driving backgrounds. We consider AR colors to be robust if they (1) support effective visual search task performance with low variability, and (2) retain their color semantics independent of background. Twelve experienced drivers performed two types of visual tasks; a structured (text) and a semi-structured (symbol) visual search task; using a head-up display. Drivers experienced eight user interface colors against three backgrounds (brick, grass, pavement) with repetition. We measured user task performance (response time & errors) as well as color recognition (color matching and naming).

Our results showed no significant differences in response time across backgrounds, but identified significant differences in response time between colors within each background. For both text and symbol tasks, participants consistently matched perceived AR user interface colors to World Color Survey stimulus palette locations that were brighter as compared to the source AR color locations. Blue, green, and yellow AR colors showed very little hue shift, and participants verbally identified these colors by name with a high degree of accuracy as compared to other colors. Our results are generally consistent with other findings, in that we observed washout effects when using AR technology. Despite the washout, we found blue, green, and yellow AR colors to be fairly robust across all background and task conditions.

Keywords: Head-Up Display, Augmented Reality; Visual Display Image; Vehicle Display System, Color Perception.

Index terms: H.1.2 User/Machine Systems: Human Factors; H.5.m Information Interfaces and Presentation (e.g., HCI): Miscellaneous.

1 INTRODUCTION

Augmented reality (AR) is steadily moving towards mainstream commercial use, and advances in the last decade have been made in regards to vehicle AR interfaces and navigation systems in order to reduce distraction and risk while driving. Global vehicle accidents result in 1.3 million deaths and over 500 billion dollars in costs annually [1], and analysis of distracted driving found that the risk of a crash or near-crash among novice drivers increased with the performance of many secondary tasks [2]. Requiring standard new-car options such as Bluetooth or other automobile technologies that inhibit handheld cell phone use should also be considered to reduce risk of accident when driving [3]. Interfaces using augmented reality can reduce this risk, and are able to deliver information to a driver without significantly increasing mental workload [4]. For the most part, the technology needed to generate AR head-up display (HUD) interfaces is readily available [5], but needs to be further refined and integrated into the user's natural environment to realize its full potential [6]. Specifically, designing visually effective AR interfaces for use in arbitrary outdoor locations remains a challenge. As such, the success of future AR

technologies will depend on designers' understanding of AR display limitations as well as underlying user perceptual and cognitive issues.

Optical see-through AR allows users to view real world scenes and superimposed computer-generated graphics through an optical lens system (as opposed to digital video capture & presentation). Some basic advantages of displaying information in cars using AR HUD interfaces are outlined by exploratory studies [7, 8]. For example, by painting the road with semitransparent directions, navigation instructions can be visually displayed directly to the user and decisions may be color coded to warn the driver of diverging paths or turns. AR displays additionally help minimize danger resulting from drivers diverting their view as compared to traditional head-down displays [6]. For example, Medenica et al., [9] compares multiple types of navigational interfaces and found that a personal navigation device that used and AR HUD allowed for more visual attention available for the road as compared to three other tested types of personal navigation devices.

1.1 AR Color Perception

Though AR HUDs allow digital information to be delivered to drivers in a seemingly superior fashion (i.e., head-up with eyes on road), several major AR display issues can render the information unhelpful or even harmful. An early attempt to integrate AR into a gaming system noted that choosing colors to represent virtual objects in outdoor AR maps proved difficult. Some colors were difficult to distinguish from natural surroundings in direct sunlight and darker colors tended to appear translucent, becoming altered or washed out in sunny outdoor conditions [10].

Livingston, Barrow and Sibley [11], note three perceptual differences between real and graphical portions of the environment that relate to human perception: limited display resolution, lower contrast, and shifted colors. The users' experience of these differences depends on the following three aspects: the optical elements present, the display technology, and surrounding visual context [11]. These differences accumulate to distort the virtual image users perceive compared to what is originally projected (i.e., the display source image). Color blending may further be characterized by the path that light goes through to reach the users' field of view, including emission, reflection, combination with the display, and final product [12].

Though users may be able to overcome color distortions, current AR display technology does not enable users to constantly and accurately perceive color of contrast [11]. Research has found several common colors that perform poorly on all platforms and conditions, such as magenta and cyan [11], and another study noted colors to avoid are all intensities of cyan, orange, magenta, pink, and red [10].

1.2 Color Blending in Optical See-through AR

Under changing light conditions, the background textures, colors and lighting of an AR environment may also pose considerable problems [13]. Render distortion (i.e. how the display renders colors) could also have a higher impact on color blending than does the material distortion (i.e., how background colors are changed by

2016 IEEE VR 2016 Workshop on Perceptual and Cognitive Issues in AR (PERCAR)
19 March, Greenville, SC, USA
978-1-5090-0838-4/16/\$31.00 ©2016 IEEE

the display material) [14]. User perception and performance in text-based visual tasks using an optical see-through AR display also depends significantly on a complex interaction between the text color, text style, and background the text is overlaid upon [15]. Data from users' perception of color, depth, and size using AR displays is often difficult to accurately obtain and categorize. Gabbard and Swan [16] favored outdoor studies as it afforded more natural environmental cues and lighting for the application of AR, and argued that outdoor lighting creates more varied test bed since indoor illuminance only varies by 3 orders of magnitude while outdoor illuminance varies by 8 orders of magnitude. However, additional perception biases may also result from the mere presence of a HUD AR interface [17].

Some existing solutions to color blending include occlusion support and color correction. The occlusion approach uses spatial light modulators such as liquid crystals to block light coming from the background and to keep it from blending with the AR display color [18]. Sridharan et al., argued however that effective color correction depends on an accurate color blending model [14], and created binned profiles to describe color blending in optical see-through displays. Gabbard and Swan found that, when dealing with see through AR interfaces, maximizing contrast between text and background by correcting for maximum saturation and contrast was important for good text readability [15].

Despite a significant amount of study concerning color blending and text legibility in optical see-through AR, there is still a lack of research concerning which subsets of colors are most effective for both text readability and symbol recognition under multiple common backgrounds. For this work we are not interested in predicting the blended or perceived color per se or actively driving a display to produce a target blended color, as this problem space is already significantly explored [14]. Instead, we are interested in quantifying the differences in user performance and color perception as user interface elements are presented on different backgrounds when in a vehicle. In order to make the most effective use of AR HUD technology in vehicle interfaces, we must determine which subsets of colors are most robust when used in real-world driving environments. This study therefore explores the robustness of color variations under different conditions and examines emerging patterns among the robust subsets.

2 METHODS

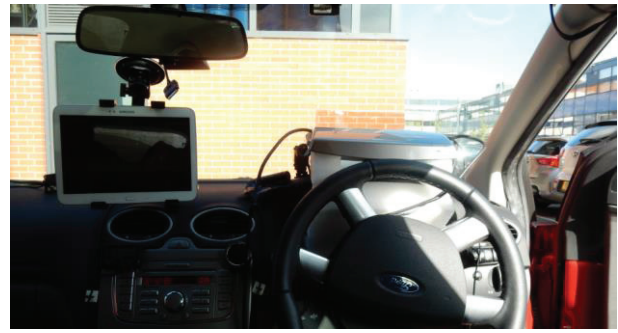
2.1 Participants

Twelve gender-balanced participants completed the series of experiments. The participants ranged from 21-44 years old with at least one year of previous driving experience in the United Kingdom and perfect or corrected vision (men: mean age =28, SD = 8; women: mean age =27, SD = 6.8).

2.2 Materials

We used Microsoft PowerPoint and Matlab software to provide visual cues to participants. We placed a Pioneer laser scanning HUD with a virtual image distance of approximately 3 meters from drivers' eyes and in line with drivers' line of vision, approximately where a user's resting gaze would be located (Figure 1).

Both the PowerPoint and Matlab programs were embedded with time tracking, and systematically controlled images displayed on the HUD to participants, while simultaneously capturing response time data. For each trial, participants mapped their perceived color on a tablet mounted to the windshield of the car. The tablet display was powered at all times (with screen saver and sleep modes disabled) to be referenced as needed when matching colors.



2.3 Figure 1: Experiment setup. The HUD was fixed to the dashboard and adjusted to the user's line of sight. A color palette was displayed using a mounted tablet situated to the left of the driver, simulating a large tablet-style in-vehicle display.

We used a text-based task as a mechanism to examine structured visual search tasks; where users employ known strategies (i.e., left-to-right, top-to-bottom) to visually examine a set of stimuli in order to complete the task. Throughout the text task, participants were instructed to find and record the number of times a target letter occurred in four lines of pseudo text. For this study, we used pseudo text in place of familiar words to minimize influence from a user's reading pace or previous knowledge of words or context. The pseudo text was built from a randomized sequence of letters, both capitalized and lowercase, spaced similarly to written words but following ISO 9241-3. To complete the text task, participants scanned the pseudo text to locate and count the occurrences of a pre-determined target letter (lower case 'e', Figure 3).

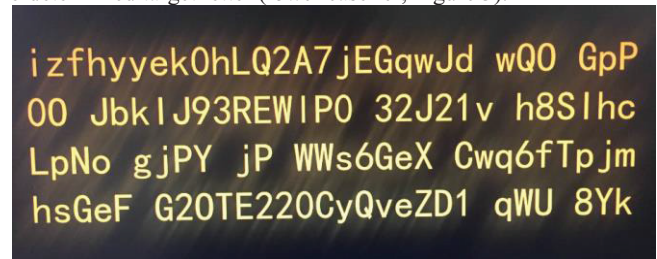


Figure 2: Our text task (a structured visual search task) required participants to identify the number of lower case 'e' characters. The correct response for this set is four.

Each trial consisted of four lines of text with at least 1 and no more than 9 target letters. Participants counted each target number aloud as identified and after reading through the text, participants immediately pressed a mouse button connected to the slideshow which recorded their completion time via the embedded software. The researcher recorded the number of targets reported by participants for that specific set of pseudo text (and experimental conditions).

2.4 Symbol Task

Participants also completed a series of semi-structured visual search tasks by identifying a target symbol within a grid of 3 by 3 symbols (Figure 3) randomized from a standardized library of commonly known symbols with similar levels of visual recognition and complexity [19]. Following completion of each grid search task, participants likewise immediately recorded their response time via mouse button press and verbally denoted the location of the target symbol. Grid location was defined simply, using the terms left-middle-right, and top-middle-bottom for x and y dimensions respectively.

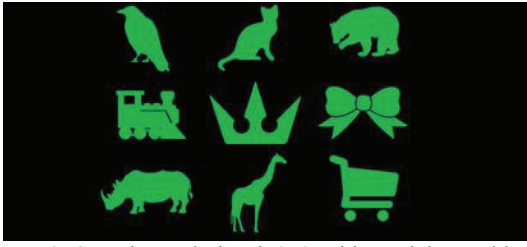


Figure 3: Sample symbol task 3x3 grid. Participants identified the correct symbol via verbal cue.

2.5 Color Matching Task

During both the text and symbol tasks, participants were given a color palette displayed on a mounted tablet showing a variety of colors ordered by hue (left to right) and brightness (top to bottom, Figure 4). Following completion each task trial (text and symbol), participants used this palette to indicate the closest AR stimuli color they perceived through the HUD. Participants verbally designated the closest color match using letters A-G for row position and numbers 1-20 for column position.

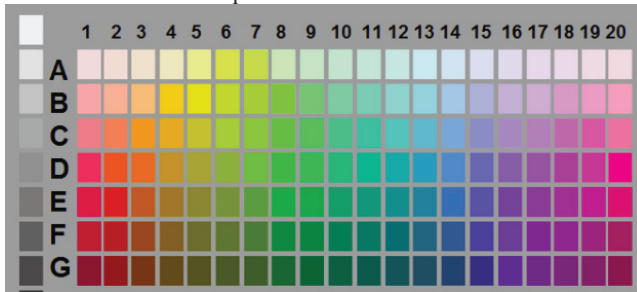


Figure 4: Truncated color palette used color matching in all tasks. The palette is created a subset of the colors from the original 330 Munsell chips from the World Color Survey.

The 170 colors used in the study were selected from the World Color Survey (WCS) stimulus palette consisting of 330 Munsell color chips [20]. We used a subset of the 330 WCS colors for two reasons: first the smaller palette was easier to visually scan given the size of our tablet; and secondly, we did not want the color matching task to be overly demanding. The truncated palette (Figure 3) uses a proportional number of the original selection by equidistant spacing. From the subset, we defined ten colors as “basic color categories” using Berlin and Kay’s process of categorizing basic color terms by their seven stage evolution process [21]. Our color choices are based on human visual perception and English cultural norms as explored by the WCS [22]. Therefore, we consider a perceptual representation of color and use the CIE color space to define chromaticity and luminance[23] (as opposed to technology-based color coding where red=255,0,0; blue=0,0,255; and so forth).

To quantify this space in relation to our WCS palette, we first measured the 1931 CIE xyY values for the 170 color chips as rendered through the mounted tablet display participants used during trials. We then identified the specific chips that corresponded to the eleven English focal responses in the Munsell color space (i.e., the centroid chip for ‘red’, the centroid chip for ‘blue’) [21]. These chip locations, and corresponding measured CIE xyY values, functioned as our primary colors displayed via the AR HUD for all tasks. We excluded black, white and gray from our selection because they rendered poorly on the Pioneer HUD, leaving eight basic colors to render on the HUD for the text and symbol tasks.

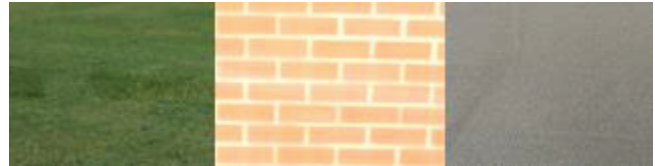
To ensure the CIE values displayed in the HUD matched the CIE values of the corresponding chips displayed via tablet, we color

matched (in the dark) the projected light through the HUD for each of the 8 colors to the xyY values viewed on the tablet to within 0.01 chromaticity distance in the CIE color space and 10 cd/m² for luminance.

2.6 Backgrounds

Each participant completed text and symbol tasks using the Pioneer head-up display against three different backgrounds; brick, pavement, and grass (Figure 5). We selected backgrounds with a high tendency to appear in vehicle operations, based on previous studies (e.g., [15]).

Figure 5: We chose each real world background based on conditions commonly encountered during vehicular operation



2.7 Location and Lighting

We conducted all trials in outdoor locations with real world backgrounds. Backgrounds were located within five minutes of walking distance of each other. We used a standard lux meter to measure ambient light before each participant began trials by reflecting light off a non-glossy white surface. We conducted testing within ambient lighting between 500-2000 lux, i.e. lighting corresponding to conditions between a dark overcast day and light overcast day. The testing took place outside since the displays are intended to be used in outside settings. This experiment was performed in {REMOVED FOR DOUBLE BLIND} because of the less varying levels of light.

2.8 Procedure

The order of backgrounds and visual search task (text; symbol) was counterbalanced to mitigate learning effects on responses. Participants sat in the driver seat and first familiarized themselves with the displays and environment in which they completed the tasks. Each participant experiences two different tasks (symbol and text) testing eight colors against three separate backgrounds, for a total of 48 conditions that were tested with four repetitions (48*4=192 total trials). A short rest interval defined by the user followed each trial, and a short travel period occurred when moving between backgrounds.

3 ANALYSES

This paper reports on dependent measures related to structured- and semi-structured tasks (response time and error), and participant verbal and matching responses of perceived colors on the displays.

3.1 Response Time

The mean response time (RT) and standard deviation for each AR source color and task type is shown in figure 5 below.

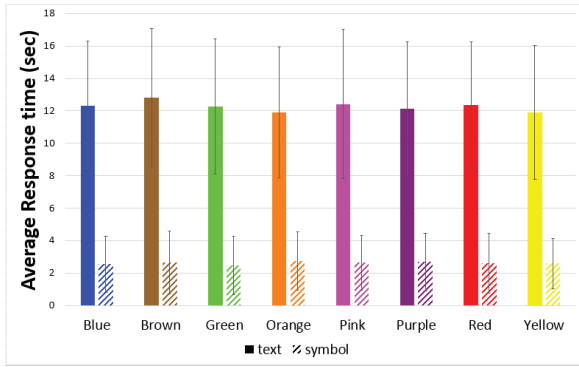


Figure 6: Mean response time and variation for blue, green, orange and yellow are slightly faster & smaller as compared to other colors.

For the purpose of this study we do not extensively compare the RT between the text and symbol tasks since they are different types of visual search tasks with different demands and resulting overall response times (text: $\mu = 12.263$ sec, $\sigma = 4.111$ sec; symbol: $\mu = 2.687$ sec, $\sigma = 1.915$ sec). Response times for both text and symbol tasks were not normally distributed, either initially or after logarithmic transformation. Therefore we used Wilcoxon signed rank tests to evaluate individual differences between colors and backgrounds. We tested each color pair across the backgrounds aggregately.

Only one significant difference (brown vs. orange) was detected when comparing color pairs.

3.2 Task Error

We categorized task error for both text and symbol tasks using a simple error metric: correct or incorrect. For the text task, incorrect counts of the letter 'e' were counted as errors. For the symbol task, incorrect assignment on the grid counted as an error. As expected, error rates for the text task (21.19%) were higher than the error rates for the symbol task (4.10%) due to the text tasks' higher level of complexity.

By background, pavement resulted in the fewest errors for the text task (17.24%) and brick for the symbol task (3.18%). Brick resulted in the most errors for both the text (25.94%) and for Grass the symbol (5.25%) tasks.

3.3 Color Matching

To qualitatively express participants' matching data, we plotted frequency of color matching responses on respective areas of our truncated WCS stimulus palette. We mapped each color matching frequency on its own frequency map, with green and purple combined in Figures 7 and 8 as examples (text and symbol tasks respectively).



Figure 7: Example contours comparing the distribution and spread of color matching responses for green and purple (text task).

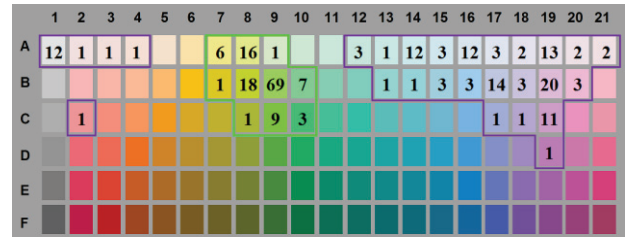


Figure 8: Example contours comparing the distribution and spread of color matching responses for green and purple (symbol task). Spread patterns were consistent by color across tasks.

All colors demonstrated a vertical shift on the WCS palette, except for brown, which was verbally identified as a dark green. For both text and symbol tasks, blue, green, and yellow displayed both the least horizontal shift and spread across total area of the map.

We additionally analyzed the color matching results using the measured CIE xyY values for each color response. Specifically, we averaged color responses data by repetition per participant, background, task type, and color. Figure 9 shows two examples of this analysis (green and purple colors in the symbol task), depicting participants' matched colors (•) as compared to the actual color that was displayed via AR HUD (x).

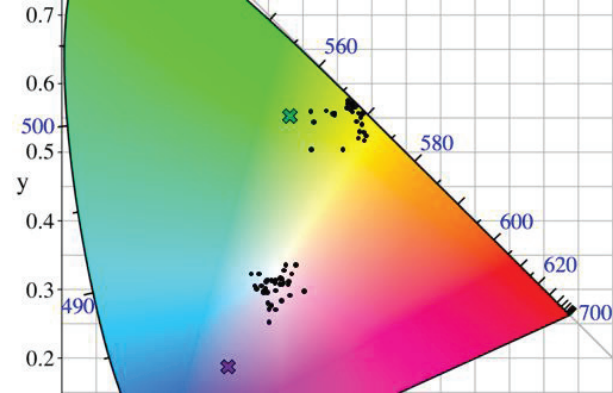


Figure 9: Example xyY CIE x and y deviation values from original defined values. Green show more uniformity and less dispersion as compared to purple.

In addition to general CIE color space mapping, we further quantified the degree of dispersion in CIE xyY space using the root mean square distance [24]. We calculated and ranked the dispersion levels for each color for both text and symbol tasks.

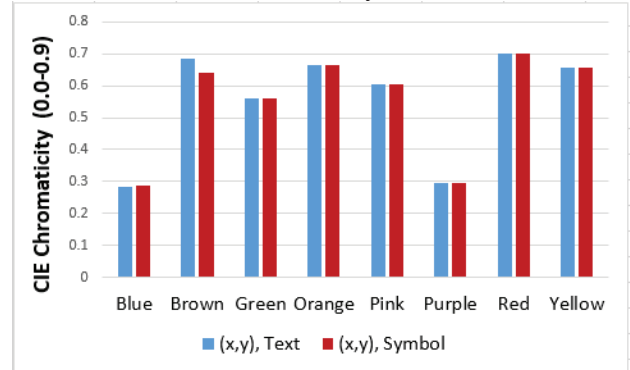


Figure 10: Chromaticity dispersion for colors within both text and symbols tasks. Higher dispersion of x and y values generally correlates inversely with name recognition.

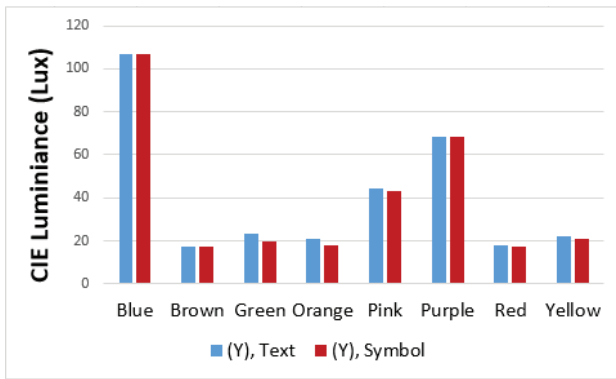


Figure 11: Luminance dispersion for colors within both text and symbols tasks. Luminance dispersion had little impact on name recognition.

3.4 Color Naming Accuracy

Because participants were allowed to be as verbally specific as desired when identifying colors, we categorized participants verbal responses based on a binary system: either included or excluded from a given color category based on whether or not the color name was used in the description. For example, “yellow-orange” would be included in the category yellow, but “cream-orange” would not (since it did not include the term “yellow”). We categorized participants’ naming accuracy for all eight colors and summarized the accuracies in Figure 12.

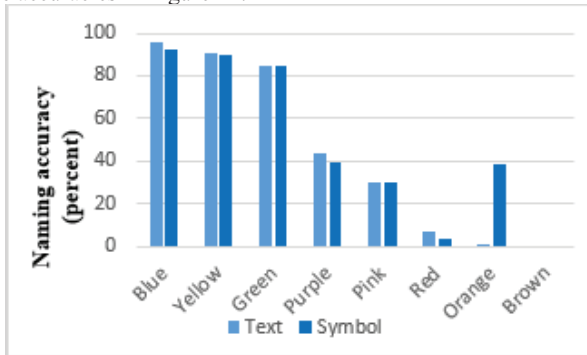


Figure 12: Blue, green and yellow retained their basic color names in over 80% of the trials in both text and symbol tasks.

Blue, green, and yellow were associated with higher accuracies for naming, while red, orange, and brown showed the least consistency and very low accuracy across all backgrounds. An interesting note is that orange showed no accuracy for naming within the text task, but showed a marked improvement for the symbol task.

4 DISCUSSION

This study seeks to better characterize the degree of robustness for different colors across background conditions when viewed through AR HUD technology. Our results indicate that when viewed through a HUD, blue, green and yellow perform more consistently and reliably as compared to other generally recognizable colors. Overall, brown performed the most poorly compared to other colors due to its low illuminance resulting in relatively higher levels of transparency when presented through our additive color-based HUD. It is interesting to note that these three colors also retained naming constancy across both text and symbol tasks. Though each color presented on the HUD suffered from noticeable brighter shifts when mapped on the WCS stimulus palette, blue, green and yellow showed very little hue shift on the

WCS or dispersion (i.e. variation) when compared to their original contour mapping in previous WCS studies [21]. These colors also demonstrate similar lower levels of dispersion when quantified within the CIE color space. Lastly, these colors consistently retained their name when verbally identified by participants, and remained recognizable despite their shift in brightness. Other colors either suffered from large dispersion when mapped on the WCS palette and/or CIE color space, and were named as entirely different colors than what was actually presented.

Similarly, participants demonstrated slightly lower response times and overall variation when performing tasks with blue, green and yellow as compared to other colors. We are surprised at the high degree of error for yellow’s performance within the text task. While yellow performed exceptionally well within the symbol task, it was associated with the most errors for the text task. We believe this large difference in performance may be due to a significant interaction between the text style, backgrounds and that particular color in a similar manner as explored previously in [15]. Regardless, low rates of RT and dispersion when color mapped still support yellow as a candidate for high robusticity.

4.1 Task design

We chose to examine two different visual search tasks in order to examine human performance on two types of cognitive processing; top down (symbol task) vs. bottom up (text task) with the added cognitive element of translating verbal cues (target symbol) into visual symbols. We expected generally lower response times for the symbol task (a qualitatively easier task to perform), however we were still surprised at the large difference in RT between task types. This likely occurred because the images used for the symbol task have a larger visual footprint as compared to a text character and thus allow for faster and more uniform processing, reducing overall differences between color pairs in symbol task response times. We attribute the lack of significant findings across color pairs in the symbol task to the lower complexity (3x3 grid) it faced due to limitations of the HUD display resolution (720x260 pixels) and resulting range for symbol grid size and arrangement within that area. The study piloted a 5x5 version of the grid for use in the symbol task, but the majority of symbols were unrecognizable due to their compressed resolution. We expect future versions of HUD display technology to support higher resolutions and allow for further inquiry into color blending and visual acuity.

Additionally, lighting played a role within the study, functioning as a source of confounding variance. Because we focused our study within real-world outdoor locations, some natural fluctuation in lighting occurred due to shifting weather patterns. Participants noted increased difficulty in completing the text task on brick when experiencing lighting fluctuations, which likely accounts for a portion of the increased error rates for text tasks on brick. AR interface colors were also affected on an individual basis by fluctuating lighting, for example green on grass and red and orange on brick were often reported to be more difficult to distinguish if illuminance as measured at the participant’s location increased to over 1000 lux.

4.2 Color and Luminance Dispersion

Within the WCS frequency and CIE color space mapping, we noticed several patterns. First, the colors which retained their names exhibited the lowest dispersion levels for chromaticity (i.e. dispersion of x and y values over the CIE color space). Conversely, colors which had higher values for dispersion could not retain their name when verbally identified. Another interesting note was that the dispersion for luminance (CIE Y) values had little effect on the accuracy to which colors were verbally identified. For example, while blue had the highest level of dispersion of luminance, it

retained high levels of naming accuracy since it was also the lowest for dispersion in chromaticity. In contrast, red had a low level of mapped dispersion for luminance (as its rendered luminance is already very low), but had a large dispersion for chromaticity and hence could not be named with accuracy. This suggests that the displayed resulting brightness of colors (after blending) does not determine the accuracy to which they are identified as much as the accurate rendering of color does.

5 FUTURE WORK

Previous studies have outlined some characteristics in color choice that promote faster cognition and accuracy in HUD technology. This study focuses on the importance of naming recognition in addition to actual measured naming accuracy when identifying and using colors within AR technology. While actual measured shifts within chromaticity and luminance may occur when rendering colors in a HUD, verbal identification of those colors are not necessarily subject to similar changes. The ability to verbally identify colors viewed through an AR HUD is not nearly as affected by luminance shifts and dispersion as that of chromaticity. We believe these results may paint a clearer picture of which colors could have a general advantage when used in AR HUD applications, and how to better track the patterns which more greatly impact their recognition across multiple types of tasks. One limitation of this study is that it being conducted while stationary meant that participants were not exposed to change in background nor motion cues while testing. Future studies could further examine and quantify this problem space by replicating similar conditions within an artificial setting such as a simulator while participants directly engage in a driving task. Creating studies with a balance of functional and environmental realism will continue to be a key issue and challenge for future studies.

6 ACKNOWLEDGEMENTS

We would like to thank {REMOVED FOR DOUBLE BLIND} for their support and assistance throughout the project, and {REMOVED FOR DOUBLE BLIND} for funding the research. We also thank {REMOVED FOR DOUBLE BLIND} for extensive help in data collection and analysis, and {REMOVED FOR DOUBLE BLIND} for her generous contribution of time and assistance during data collection.

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